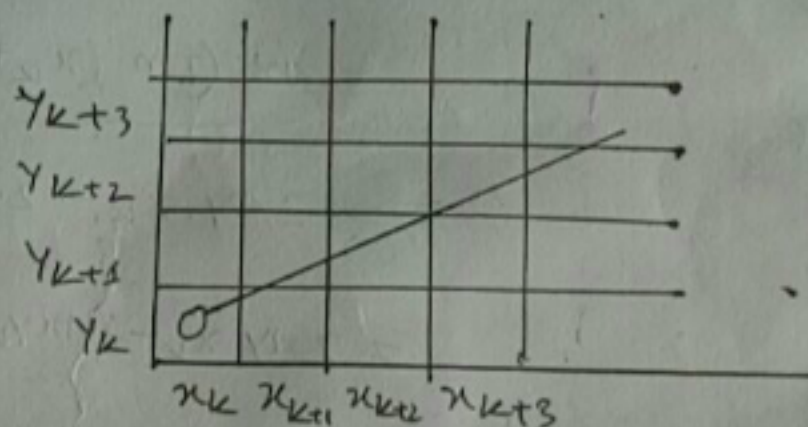
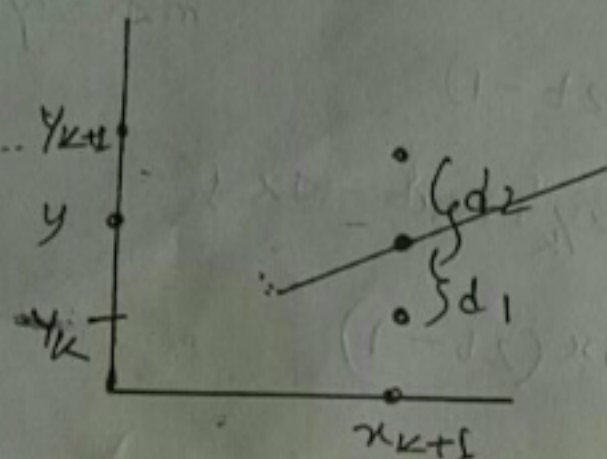


Bresenham's line Algorithm

Assuming we have determined that the pixel at (x_k, y_k) is to be displayed. We next need to decide which pixel to plot in column x_{k+1} . Our choices are the pixels at positions (x_{k+1}, y_k) and (x_{k+1}, y_{k+1}) .



At sampling position x_{k+1} , we label vertical pixel separations from the mathematical line paths as d_1 and d_2 . The y coordinate on the mathematical line at pixel column position x_{k+1} is calculated as,

$$y = m(x_{k+1}) + b$$

Then,

$$d_1 = y - y_k$$

$$= m(x_{k+1}) + b - y_k$$

and

$$d_2 = (y_{k+1}) - y$$

$$= y_{k+1} - m(x_{k+1}) - b$$

The difference between these two separations is,

$$d_1 - d_2 = 2m(x_k + 1) - 2y_k + 2b - 1$$

A decision parameter P_k for the k th step $\rightarrow$

$$P_k = \Delta x (d_1 - d_2)$$

$$m \Delta x = \Delta y$$

$$= \Delta x (2m(x_k + 1) - 2y_k + 2b - 1)$$

$$= 2\Delta y \cdot x_k - 2\Delta x y_k - 2\Delta x y_k + 2\Delta x b - \Delta x$$

$$= 2\Delta y \cdot x_k - 2\Delta x y_k + 2\Delta y + \Delta x (2b - 1)$$

$$= 2\Delta y \cdot x_k - 2\Delta x y_k + C$$

At step $k+1$ the decision parameter is evaluated.

$$P_{k+1} = 2\Delta y \cdot x_{k+1} - 2\Delta x \cdot y_{k+1} + C$$

$$\therefore P_{k+1} - P_k = 2\Delta y (x_{k+1} - x_k) - 2\Delta x (y_{k+1} - y_k)$$

But $x_{k+1} = x_k + 1$, so that.

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x (y_{k+1} - y_k)$$

where the term $y_{k+1} - y_k$ is either 0 or 1, depending on the sign of parameter P_k .

The first parameter, P_0 , at the starting pixel position (x_0, y_0) and $m = \frac{\Delta y}{\Delta x}$

$$P_0 = 2\Delta y - \Delta x.$$

Algorithm:

1. Input the two line endpoints and store the left endpoint in (x_0, y_0)

2. Load (x_0, y_0) into the frame buffer; that is plot the first point.

3. calculate constants Δx , Δy , $2\Delta y$, and $2\Delta y - 2\Delta x$ and obtain the starting value for the decision parameter as

$$P_0 = 2\Delta y - \Delta x$$

4. At each x_k along the line, starting at $k=0$ perform the following test

if $P_k < 0$, the next point to plot is (x_{k+1}, y_k) and ~~then~~

$$P_{k+1} = P_k + 2\Delta y$$

otherwise, the next point to plot is (x_{k+1}, y_{k+1}) and

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x$$

5. Repeat step- 4 Δx times.

Example: The endpoints $(20, 10)$ and $(30, 18)$. The slope of 0.8. with $\Delta x = 10$, $\Delta y = 8$

The initial decision parameter has the value

$$P_0 = 2\Delta y - \Delta x$$

$$= 6$$

$$2\Delta y = 16, \quad 2\Delta y - 2\Delta x = -4$$

k	P_k	x_{k+1}, y_{k+1}	k	P_k	x_{k+1}, y_{k+1}
0	6	(21, 11)	5	6	(26, 15)
1	2	(22, 12)	6	2	(27, 16)
2	-2	(23, 14)	7	-2	(28, 16)
3	14	(24, 13)	8	14	(29, 17)
4	16	(25, 14)	9	10	(30, 18)

